Data/Math (2026), **xx**:xx 1–13

CAMBRIDGE
UNIVERSITY PRESS

Data Paper

SynopticBench: Evaluating Vision-Language Models on Generating Weather Forecast Discussions of the Future

Timothy B. Higgins[1,2], Antonios Mamalakis[1,2] and Chirag Agarwal[2]

[1]Department of Environmental Sciences, University of Virginia, Charlottesville, 22903, Virginia, United States.
[2]School of Data Science, University of Virginia, Charlottesville, 22903, Virginia, United States.

Keywords: Multimodality, Numerical Weather Prediction, Vision Language Model, Weather Forecasting

MSC Codes: *Primary* – CODE1; *Secondary* – CODE2, CODE3

Abstract
Recent advances in visual-language models (VLMs) have led to significant improvements in a plethora of complex multimodal tasks like image captioning, report generation, and visual perception. However, generating text from meteorological data is highly challenging because the atmosphere is a chaotic system that is rapidly changing at various spatial and temporal scales. Given the complexity of atmospheric phenomena, it is critical to verifiably quantify the effectiveness of existing VLMs on weather forecasting data. In this work, we present SynopticBench, a high-quality dataset consisting of 1,367,041 text samples of Area Forecast Discussions created by the National Weather Service over the continental United States paired to images of 500mb geopotential height, 2 meter temperature, and 850mb wind velocity in weather forecasts. We also present Synoptic Phenomena Alignment and Coverage Evaluation (Space), a novel evaluation framework that can be used to effectively estimate the quality of text descriptions of synoptic weather phenomena. Extensive experiments on generating forecast discussions using state-of-the-art VLMs show the sensitivity of existing evaluation metrics in this domain and enable further exploration into synoptic weather and climate text generation.

Impact Statement

To the best of our knowledge, this is the first study to generate text discussions from images of numerical weather forecasts. We introduce i) the largest benchmark dataset, comprising 1,367,041 image-text samples, to run experiments that explore the potential to generate text discussions from numerical weather prediction; and ii) a methodology for evaluating generated text discussions of synoptic weather and climate.

Contents

1 Introduction **2**
2 SynopticBench **3**
2.1 Dataset Building Pipeline . . . 3
2.2 Dataset Collection . . . 3
2.3 Preprocessing . . . 4
2.4 Baselines . . . 4
2.5 Synoptic Phenomena Alignment and Coverage Evaluation (Space) metric . . . 5
3 Results **6**
4 Conclusion **8**

A Experimental details **11**

1. Introduction

Recent advances in vision-language models (VLMs) have enabled substantial progress on complex multimodal tasks such as image captioning, document understanding, visual question answering, and multimodal summarization. As these models improve in capability and are increasingly deployed or proposed for deployment in high-stakes settings, there is a growing need for evaluation frameworks that go beyond surface-level text similarity and instead quantify whether generated language is faithful, spatially grounded, and domain-relevant.

To this end, weather forecasting is a particularly consequential and technically challenging high-stakes domain. Numerical Weather Prediction (NWP) is deeply integrated into business operations, agriculture, government decision-making, public safety, and daily life for billions of people [8, 7, 15, 16, 2]. Vast observation networks and reanalysis products have enabled continual improvements in model development, data assimilation, and assessments of predictive reliability [6, 13]. However, atmospheric dynamics are chaotic and span a wide range of spatial and temporal scales, which is why forecast verification relies on many complementary skill metrics (*e.g.,* Anomaly Correlation Coefficient, RMSE, MAE, and skill scores) to characterize different aspects of performance [3]. In contrast, despite the operational importance of narrative forecasting products, methods to verifiably quantify the reliability of textual representations of atmospheric states and evolution **remain limited**.

Recent years have seen a growing body of work that explores multimodal learning for atmospheric data, including fine-tuning frontier VLMs and dataset creation for meteorological question answering (QA) and event understanding. Examples include CLLMate [9], which pairs extreme-event news with reanalysis for QA; RadarQA [5], which introduces radar annotations and heuristics for VLM fine-tuning; and hazard-focused datasets such as SEVIR [18] and GridRad-Severe [11]. Additional efforts pair structured meteorological inputs with textual outputs for QA-oriented benchmarks, such as Meteor-Pred [14], Climate IQA [4], Zephyrus [17], and WeatherQA [10]. While these studies demonstrate the promise of multimodal models for atmospheric applications, direct comparisons across benchmarks are often confounded by rigid constraints on target outputs, spatial scales, forecast horizons, and training data choices. Moreover, many evaluations emphasize question-answering accuracy or rely heavily on traditional text metrics (Bleu, Rouge, Meteor, BertScore) and/or LLM-judge scoring, which can be insufficient for assessing whether generated discussions correctly identify synoptic phenomena, capture their spatial extent, and reflect scale-dependent relevance to a forecast domain.

Present work. To address all the aforementioned gaps, we introduce a new benchmark and evaluation methodology designed specifically for the generation and verification of synoptic-scale forecast discussions. We present SynopticBench, a large-scale data set comprising 1,367,041 National Weather Service Area Forecast Discussions paired with forecast images of 500mb geopotential height, 2m temperature and 850mb wind velocity over the continental United States. In contrast to existing benchmarks, SynopticBench is the first benchmark to evaluate NWP meteorological samples across both mesoscale and synoptic spatial scales for all weather (see Table A). Critically, we also propose Synoptic Phenomena Alignment and Coverage Evaluation (Space), an evaluation framework that explicitly accounts for the scale-dependent nature of atmospheric phenomena and the differing spatial precision required across variables (*e.g.,* precipitation versus pressure patterns). Space is designed to evaluate distinct phenomena separately while incorporating spatial coverage and relevance, enabling more faithful estimation of a VLM's skill in generating meteorological text. Building on SynopticBench, we train models for synoptic discussion generation and fine-tune widely used VLMs. Our results show that the ability of the models to conduct physical reasoning is improved from finetuning. The key goal of SynopticBench is to advance the study and understanding of vision-language models in weather forecasting. Therefore, we

Dataset	Meteorological Data	Text Size	Spatial Scale	Target
GridRad-Severe [11]	RS	x	SC, MC, MS	Extreme Events
SEVIR [18]	RS	x	MS	Extreme Events
WeatherQA [10]	Reanalysis	8,000	MS	Extreme Events
ClimateIQA [4]	Reanalysis	762,120	MS, Synoptic	Extreme Events
CLLMate [9]	Reanalysis	41,000	MS	Extreme Events
RadarQA [5]	RS	69,000	Mesoscale	All Weather
MP-Bench [14]	Reanalysis	421,363	MS	Extreme Events
Zephyrus Bench [17]	Reanalysis	2,158	MS, Synoptic, Global	All Weather
SynopticBench	**NWP**	**1,367,041**	**MS and Synoptic**	**All Weather**

Table 1. *Comparison between SynopticBench and existing work using multimodal datasets for atmospheric data. Here, we present the type of meteorological data (RS→Remote Sensing; NWP→Numerical Weather Prediction), the number of text samples, the relevant spatial scales (SC→Single-cell; MC→Multi-cell; MS→Mesoscale), and the target of each study. SynopticBench is the first large-scale benchmark that generates text from NWP and contains 1,367,041 text samples .*

share the dataset, metric, and fine-tuned weights of a range of VLMs to support reproducible research and more rigorous, domain-appropriate evaluation of VLMs in weather forecasting.

2. SynopticBench

Here, we detail the dataset building pipeline (Sec. 2.1), data collection (Sec. 2.2), data preprocessing (Sec. 2.3), and our proposed Space metric (Sec. 2.5).

2.1. Dataset Building Pipeline

One of the largest bottlenecks that stunts progress for multimodal models for atmospheric data is the scarcity of high-quality text that can be paired with numerical data. We propose using Area Forecast Discussions (AFDs) created by the National Weather Service (NWS) as a potential solution to the multimodal data problem. The AFD text data are publicly available and were extracted from the Iowa Environmental Mesonet[1]. AFDs exist at NWS stations across the United States and are typically issued multiple times per day. They reliably describe weather forecasts to an exceptionally high degree of detail, often informing the public about the weather a week in advance [1]. In addition, they describe numerous physical variables, vertical levels, spatial scales, and processes. AFDs also typically include high-quality descriptions of the temporal evolution of weather phenomena, creating potential for training models to generate text that describes how atmospheric data changes over time.

2.2. Dataset Collection

We create SynopticBench by pairing the National Centers for Environmental Prediction (NCEP) Global Forecast System (GFS) model [12] to AFDs. This data set contains GFS forecasts issued four times per day from January 1, 2016→December 31, 2025 (14,159 total forecasts) paired with **1,367,041** AFD text samples. Each GFS forecast consists of 2m temperature, precipitation rate, 500mb geopotential height, 850mb zonal wind, and 850 mb meridional wind is matched to the nearest AFD in time (always

[1] https://mesonet.agron.iastate.edu/wx/afos/list.phtml

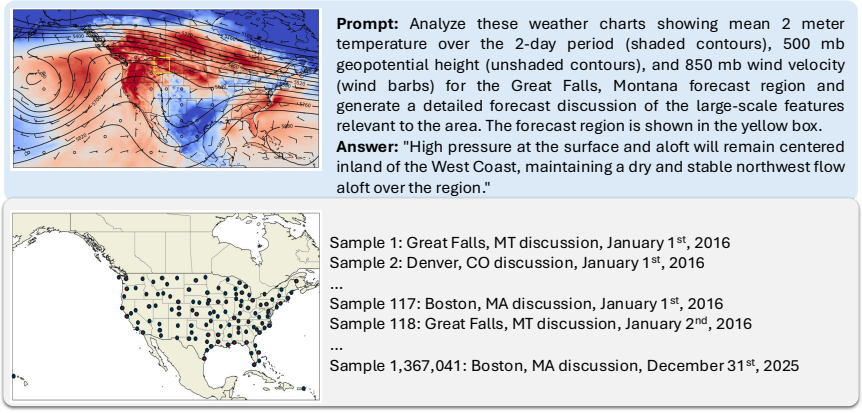


Figure 1. *An example case of a single sample from the training set (top panel). Each training sample image has a yellow box indicating the location of the discussion. The example answer is a filtered AFD. All of the locations used in the discussions are shown in the bottom panel. The format of the training samples is also shown, with 117 AFDs matched to each forecast.*

within several hours) for each location. We pair each forecast with 117 text samples, as there are 117 NWS station locations used in the study (Figure 1). This creates consistency among the text samples for descriptions of the primary synoptic features occurring over the US for a given forecast.

2.3. Preprocessing

Before we begin training and evaluating vision-language models, we apply several data preprocessing steps to both AFDs and GFS forecasts. During preprocessing, we first discard all lead times beyond 48 hours. Although AFDs primarily align with GFS forecasts, they are sometimes based on other models as well, which diverge as the lead time increases due to the chaotic nature of the atmosphere. The spatial domain is cropped from 15°N to 65°N and 60°W to 160°W. We plot mean 2-meter temperature anomalies in shaded contours along with mean 500mb geopotential height in line contours and 850mb wind velocity in the form of wind barbs. A yellow $5^\circ \times 5^\circ$ box is centered on the location of each discussion that is paired to an image.

Keeping the goal of matching text that can be represented in the images, we preprocessed the information in the AFDs for the experiments. AFDs are often informed by GFS forecasts, but can use information from a variety of different models. We remove sentences including any reference to other forecast models or synoptic pressure systems that do not exist at the scale or vertical level that the images show. We also only use sentences including keywords that do describe synoptic pressure systems and temperature anomalies including: *trough, ridge, low pressure, high pressure, cold front, and warm front*. The full list of inclusion and exclusion keywords is included in the appendix. When evaluating synoptic phenomena with lead times up to 48 hours, we can also expect robust consistency of large-scale patterns. The information described in AFDs occurs chronologically, often including the day of the week on which different phenomena occur. Once a day of the week that is more than two days after the issue day is mentioned, all text that follows is removed to stay consistent with the forecast lead times.

2.4. Baselines

We used four different baselines to add perspective to the results. Gemini-3.1 Pro is used as a baseline for current state-of-the-art models. The nearest neighbor baseline created by first computing the SSIM

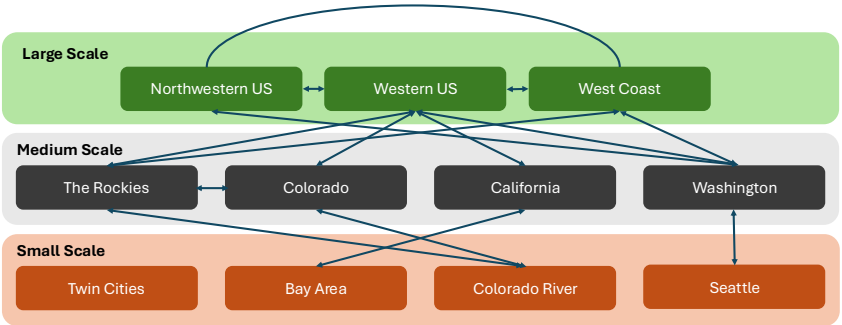


Figure 2. *Several examples of matching large- (green), medium- (purple), and small-scale (orange) location keywords. Blue lines indicate the potential matches that these locations would make if found in the predicted or reference text.*

scores between each test set image and all training set images. The text paired with the training set image with the highest SSIM score is used as the text for each sample in the baseline. The climatology is created by choosing text from a random sample in the training set from the same month and location as each sample in the test set. The blind LLM baseline strips the image tokens from test before running LLaMA-3.2-11B.

2.5. Synoptic Phenomena Alignment and Coverage Evaluation (Space) metric

Traditional metrics for text evaluation are not sufficient for text describing atmospheric processes because they do not include any physical reasoning. The purpose of Space is to evaluate the ability of the VLM to generate text that predicts the correct phenomenon in the target location. Evaluating location accuracy from text can be a complex task because many terms may differ entirely, but represent similar locations (*e.g.,* The Rockies vs. Colorado) and the precision of location terms may be more or less acceptable depending on the scale of the phenomenon (*e.g.,* "trough moving through the Denver area" vs "trough moving through Colorado" is preferred more than "rain in the Denver area" vs. "rain in Colorado". To accommodate these relations, we introduce a spatial hierarchy tree specifically created for evaluating text describing synoptic phenomena. The hierarchy comprises three levels that capture terms commonly used to detail the locations of synoptic weather phenomena in North America (see Fig. 2 for an example of a small portion of the hierarchy tree).

Space can be applied towards various types of synoptic weather phenomena that have positive and negative phase categorizations (*e.g.,* high/low pressure systems, warm/cold fronts, wet/dry events). Next, to demonstrate the methodology of Space, we use pressure systems as an example. We begin Space by matching pressure terms to locations. If 'low pressure', 'high pressure', 'trough' (excluding instances of 'shortwave trough'), or 'ridge' exist in the text, the term is first matched to any location that follows in the sentence. For a sample of text, this creates a list of pressure systems matched to different locations. Pressure terms are then matched to other instances for locations that are equal or related (see connections in Figure 2). Locations with a common relative can also match with the exception of those that cover a large portion of the continent (see Appendix). This logic creates groups of pressure objects for both the generated and the AFD text sample. Objects can be grouped for single discussions (Space-local) or aggregated across all locations (Space-aggregate) for a single forecast. If any object from a group has the same pressure sign and a location related to any object in the corresponding text sample, the object groups form a match. We consider 'trough' to be synonymous with 'low pressure', and 'ridge' to be synonymous with 'high pressure'. The match score, s_m, is a measure of the quality of the matches and

Model	Bertscore	ROUGE-L	METEOR	F1	Gemini-2.5-Flash
LLaVA-v1.5-7B-base	.7143$_{\pm.0001}$	.1091$_{\pm.0001}$	.1327$_{\pm.0001}$	.1471$_{\pm.0002}$	.1565$_{\pm.0031}$
LLaVA-v1.5-13B-base	.7192$_{\pm.0001}$	.1086$_{\pm.0001}$	.1376$_{\pm.0001}$	.1498$_{\pm.0002}$	.1896$_{\pm.0032}$
LLaMA-3.2-11B-Vision-base	.7051$_{\pm.0001}$	.1043$_{\pm.0002}$	.1212$_{\pm.0002}$	.1313$_{\pm.0002}$	.2623$_{\pm.0041}$
Qwen2.5-VL-7B-base	.7136$_{\pm.0001}$	.1083$_{\pm.0001}$	.1447$_{\pm.0001}$	.1591$_{\pm.0003}$	.2566$_{\pm.0035}$
LLaVA-v1.5-7B-LoRA	.7814$_{\pm.0002}$	.1629$_{\pm.0003}$	.1353$_{\pm.0003}$	.1943$_{\pm.0004}$	.2426$_{\pm.0048}$
LLaVA-v1.5-13B-LoRA	.7858$_{\pm.0002}$	.1663$_{\pm.0003}$	.1407$_{\pm.0003}$	.2010$_{\pm.0004}$	.2472$_{\pm.0052}$
LLaMA-3.2-11B-Vision-LoRA	.7773$_{\pm.0001}$	.1530$_{\pm.0003}$	.1174$_{\pm.0003}$	.1867$_{\pm.0003}$	.2757$_{\pm.0047}$
Qwen2.5-VL-7B-LoRA	**.7927**$_{\pm.0002}$	**.1681**$_{\pm.0003}$	.1601$_{\pm.0004}$	**.2184**$_{\pm.0004}$	.2824$_{\pm.0053}$
Gemini-3.1-Pro	.6913$_{\pm.0004}$	.0703$_{\pm.0005}$	.1456$_{\pm.0008}$	.1030$_{\pm.0009}$	**.4185**$_{\pm.0071}$
Nearest Neighbor	.7662$_{\pm.0001}$	.1194$_{\pm.0002}$	.1395$_{\pm.0002}$	.1748$_{\pm.0003}$	.1785$_{\pm.0047}$
Climatology	.7776$_{\pm.0001}$	.1434$_{\pm.0002}$	**.1632**$_{\pm.0003}$	.2035$_{\pm.0003}$	.2358$_{\pm.0049}$
LLama-3.2-11B (blind)	.7257$_{\pm.0001}$	.1101$_{\pm.0002}$	.1503$_{\pm.0002}$	.1544$_{\pm.0003}$	.2574$_{\pm.0054}$

Table 2. *Traditional metrics for base models, finetuned models, and baselines are shown. We calculate the mean and standard error of the mean Bertscore, ROUGE-L, METEOR, F1, and LLM-judge scores across samples within the test set. The range of possible scores for all metrics is 0 to 1, with 1 being a perfect score.*

is defined below:

$$s_m = 1 - \left| \frac{m_L^{\text{pred}}}{m_L^{\text{pred}} + m_H^{\text{pred}}} - \frac{m_L^{\text{ref}}}{m_L^{\text{ref}} + m_H^{\text{ref}}} \right|, \tag{2.1}$$

where m_L^{pred} is the number of matching *negative phase* objects in the predicted text, m_L^{ref} is the number of matching *negative phase* objects in the reference text, m_H^{pred} is the number of matching *positive phase* objects in the predicted text, and m_H^{ref} is the number of matching *positive phase* objects in the reference text. The coverage ratio, r_c, is the percentage of pressure objects that are included in matches and is defined as:

$$r_c = \frac{\sum((m_L^{\text{pred}} + m_H^{\text{pred}}) + (m_L^{\text{ref}} + m_H^{\text{ref}}))}{n_L + n_H}, \tag{2.2}$$

where n_L is the total number of *negative phase* objects and n_H is the total number of *positive phase* objects in the combined predicted and observed text. The coverage ratio is designed to penalize the generated text for creating hallucinations and for failing to mention key terms. The final Space score is calculated as the product of the matching score and coverage ratio, *i.e.,* $s = s_m \cdot r_c$.

3. Results

Traditional Metrics are not Sufficient. Base model versions for LLaVA-v1.5-7B, LLaVA-v1.5-13B, Qwen2-VL-7B, and LLaMA-3.2-11B-Vision were run on the test set. We first evaluate each of the four base models, finetuned models, and baselines on traditional skill metrics (Table 2). LoRA finetuning improved performance on all traditional metrics except METEOR for LLaMA-3.2-11B-Vision. The climatology baseline had the highest METEOR score, while finetuned Qwen2.5-VL-7B had the highest Bertscore, ROUGE-L, and F1 scores. Gemini-3.1-Pro had the lowest Bertscore, ROUGE-L score, METEOR score, and F1 score while having the highest LLM-as-a-judge score. The climatology baseline used text from real NWS AFDs issued from the same exact station, which may have allowed it to have relatively high scores for traditional metrics by using similar discussion styles. Despite slight variations

Model	s_s^{loc}	s_m^{loc}	r_c^{loc}	s_s^{agg}	s_m^{agg}	r_c^{agg}
LLaVA-v1.5-7B-base	.0976$_{\pm.0008}$	.4592$_{\pm.0056}$	.2158$_{\pm.0013}$	.4177$_{\pm.0008}$	.5039$_{\pm.0056}$	.9467$_{\pm.0013}$
LLaVA-v1.5-13B-base	.1553$_{\pm.0007}$	.4404$_{\pm.0061}$	.3615$_{\pm.0012}$	.4698$_{\pm.0007}$	.5044$_{\pm.0061}$	.9317$_{\pm.0012}$
LLaMA-3.2-11B-Vision-base	.1194$_{\pm.0008}$	.6296$_{\pm.00013}$	.1936$_{\pm.0012}$	.6492$_{\pm.0008}$	.6690$_{\pm.0013}$	.9648$_{\pm.0012}$
Qwen2.5-VL-7B-base	.0985$_{\pm.0018}$	.5072$_{\pm.0032}$	.1950$_{\pm.0031}$	.5831$_{\pm.0018}$	.6221$_{\pm.0032}$	.9279$_{\pm.0031}$
LLaVA-v1.5-7B-LoRA	.2235$_{\pm.0015}$	.6575$_{\pm.0023}$	.3364$_{\pm.0018}$	.6486$_{\pm.0015}$	.6714$_{\pm.0023}$	.9635$_{\pm.0018}$
LLaVA-v1.5-13B-LoRA	.2604$_{\pm.0015}$	.6601$_{\pm.0021}$	**.3900**$_{\pm.0018}$	.6410$_{\pm.0015}$	.6629$_{\pm.0021}$	.9648$_{\pm.0018}$
LLaMA-3.2-11B-Vision-LoRA	.2424$_{\pm.0016}$	.6554$_{\pm.0023}$	.3654$_{\pm.0019}$	.6091$_{\pm.0016}$	.6397$_{\pm.0023}$	.9488$_{\pm.0019}$
Qwen2.5-VL-7B-LoRA	**.2626**$_{\pm.0015}$	**.6744**$_{\pm.0020}$	.3845$_{\pm.0018}$	.6294$_{\pm.0015}$	.6498$_{\pm.0020}$	.9663$_{\pm.0018}$
Gemini-3.1-Pro	.2005$_{\pm.0037}$	.5825$_{\pm.0034}$	.3445$_{\pm.0058}$	.6685$_{\pm.0037}$	.6775$_{\pm.0034}$	**.9864**$_{\pm.0058}$
Nearest Neighbor	.1569$_{\pm.0010}$	.5485$_{\pm.0015}$	.2866$_{\pm.0015}$	.6187$_{\pm.0010}$	.6393$_{\pm.0015}$	.9661$_{\pm.0015}$
Climatology	.2009$_{\pm.0016}$	.5655$_{\pm.0012}$	.3554$_{\pm.0018}$	**.6808**$_{\pm.0016}$	**.6900**$_{\pm.0012}$	.9856$_{\pm.0018}$
Llama-3.2 (blind)	.1745$_{\pm.0010}$	.4967$_{\pm.0014}$	.3542$_{\pm.0017}$	.5690$_{\pm.0010}$	.5794$_{\pm.0014}$	.9813$_{\pm.0017}$

Table 3. *SPACE was used for synoptic pressure system evaluation for base models, finetuned models, and baselines. We calculate the mean and standard error of the SPACE-local and SPACE-aggregate scores (s_s), match scores (s_m), and coverage ratios (r_c) within the test set. The range of possible scores for all metrics is 0 to 1, with 1 being a perfect score.*

in the performance of all models, all scores were low, indicating that the language in the generated discussions had little overlap with the NWS discussions. The traditional scores cannot determine the models' ability to discuss the primary relevant features in the images, which creates a need for further evaluation.

Space scores can Reduce Evaluation Uncertainty. Although Space could be used for various types of weather phenomena, we demonstrate the utility of Space on pressure systems in this study. Here, we show how Space scores help can the user understand how each model performs in identifying the correct relevant pressure systems for the area and referring to them in the correct locations. Space-aggregate uses all locations at a given time to create a large sample size of pressure objects matched to specific locations for evaluation. It is useful for understanding the model's ability to determine the general large-scale pressure features that occur in the forecast. Space-local uses a single location at a given time to create a much smaller sample size of pressure objects but can be useful for understanding the model's ability to discuss the features impacting a specific location. All Space scores for pressure systems are shown in Table 3. Space-aggregate scores are considerably higher than Space-local scores because the increased sample size of text increases the chances that relevant locations for synoptic phenomena are mentioned in both the generated text and the AFDs. Theoretically, a model that always randomly guesses between high pressure and low pressure would have a match score of roughly 0.5 without any knowledge of climatology, which indicates that the finetuned models likely all have some skill in distinguishing between high and low pressure. The local coverage ratios (r_c) are lower than the match scores for every model, demonstrating that predicting the phenomena in the correct location is a more challenging task for these VLMs than accurately describing the correct polarity of synoptic phenomena.

The Qwen2.5-VL-7B fine-tuned model had the highest Space-local scores while Climatology had the highest Space-aggregate scores. The fine-tuned models had higher Space-local scores than all of the base models and baselines. The Space-aggregate scores for the fine-tuned models were generally higher than those of the base models with the exception of LLaMA-3.2-11B-Vision. The improvement of Space-aggregate scores over base models is impacted by both improved coverage ratios and match scores. The Space scores often agree with the traditional metrics, with the fine-tuned Qwen-VL-7B configuration generally having the highest scores. One clear exception is Gemini-3.1-Pro, which has some of the lowest scores when evaluated on traditional metrics and fairly high Space scores. The Space scores and traditional metrics also suggest that increasing the parameters from 7 billion to 13 billion

in LLaVA does not drastically improve generated discussion quality and the fine-tuning approach is far more effective.

Case Studies Illustrate the Advantages of Space. An example of the evaluation of both Space and traditional metrics in two different cases is shown in Figure 3. In Case 1, there is a high pressure system over the southwestern United States, which is the most influential pressure system relevant to the station location (Tuscon, Arizona). The NWS discussion mentions high pressure in the region, and the generated VLM text also discusses high pressure systems in the same region. This leads to a perfect Space-local score for pressure systems. The traditional metrics for this case were generally close to average for this model. In Case 2, there is a high pressure system over much of the southeastern US, which is the most influential pressure system to the station location (Columbia, South Carolina). The NWS discussion mentions a ridge in the forecast area while the VLM's generated text discusses a trough in the southeastern United States. This leads to a Space score of 0. Despite having a Space-local score of 0, the traditional metrics were unable to show a clear difference in the quality of both cases. Case 2 also shows some instances of hallucinations in which the model discusses phenomena that it could not possibly see based on the image (e.g. 1 inch of snow), which does not penalize the Space score because Space only evaluates one type of phenomenon at a time. This case study demonstrates a clear example of the reliability of Space scores relative to traditional metrics for the evaluation of individual types of phenomena in weather discussions. Several additional cases are shown in the Appendix.

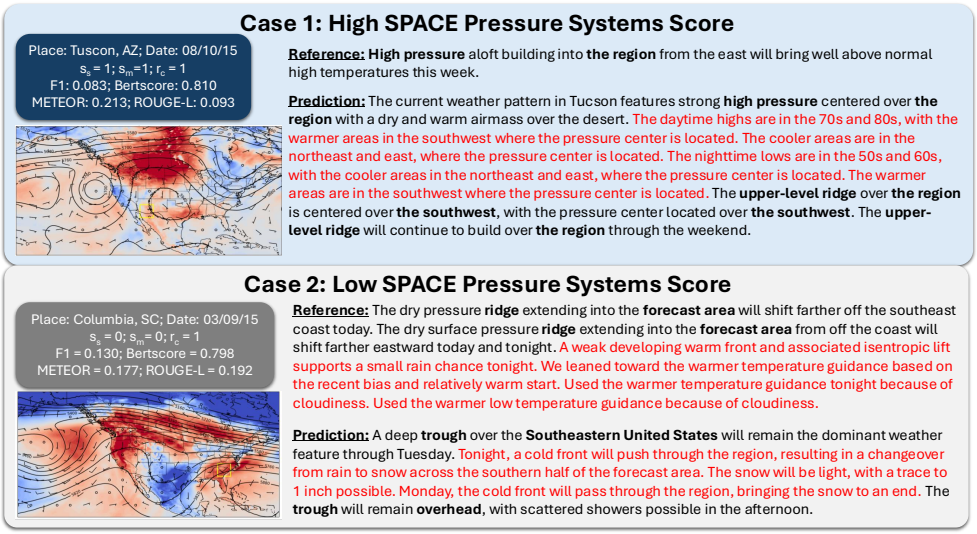


Figure 3. *Two cases demonstrating differences between Space scores and traditional skill metrics. The reference text samples are filtered NWS AFDs and the prediction text samples are generated from the finetuned version of LLaVA-v1.5-7B. The terms in bold are used to compute Space scores for pressure systems. Sentences that are irrelevant to the Space scores are shown in red.*

4. Conclusion

In this work, we introduce a dataset with more than one million samples that we use to train a Vision Language Model for discussion generation from images of weather forecasts across North America. We also introduce a skill metric (Space) to evaluate the ability of language models to identify important synoptic features in the correct locations. Fine-tuning VLM projector weights with LoRA adapters leads

to greater overall performance than the base models on both SPACE and traditional metrics. Climatology and finetuned Qwen2.5-VL-7B had the highest scores in most of the metrics in this study, with Qwen2.5-VL-7B excelling more in describing features relevant to a precise location. The failure of any model to score higher than climatology in the SPACE-aggregate scores demonstrates a key limiting factor in the ability of the models in this study to accurately describe synoptic features throughout the entire system.

The data set used in this study pairs highly complex forecast discussions from the National Weather Service with images of weather forecasts. This study performed experiments on synoptic pressure systems as an example use case and focused entirely on the United States, which creates an opportunity for future work. Human expert evaluation for text describing various types of atmospheric phenomena would also be valuable for strengthening confidence in any current or future metrics. The size and complexity of the data set allow the user to perform experiments on other types of phenomena, specific locations or regions, spatial scales, and temporal ranges. Although the finetuned models often outperformed the base models on SPACE scores, improved model architectures and loss functions, agentic frameworks, and video-language models could add additional benefits.

Weather forecasts are highly complex, and complete descriptions of them therefore often require long and detailed discussions. None of the models used in this study come close to rivaling the reliability nor complexity of NWS AFDs, so the evaluation of text generation for weather and climate applications must begin with basic isolated tasks. SPACE scores excel in capturing similarities in some of the most important details of text samples. They can help evaluate the ability of the models to demonstrate physical understanding of atmospheric processes rather than only learning common textual patterns. By determining the ability of the model to do so, the user can better understand when to stop training, which can help prevent overfitting.

Although SPACE scores are a useful method for evaluating text for multimodal language models for weather and climate applications, they do not fully capture the level of quality in generated forecast discussions, and additional metrics that evaluate text would help create a more comprehensive evaluation framework. Future improvements could involve temporal accuracy (more relevant for longer lead times), system evolution (more relevant for models that can handle videos), and vertical levels (more relevant for models that can handle more channels).

Acknowledgments

The authors thank the reviewers of this work for their dedication towards supporting the research community and providing comments that were critical towards improving the manuscript.

Funding Statement

This work was funded by the University of Virginia Environmental Institute Climate Fellows Program Grant DN002057.

Competing Interests

None

Data Availability Statement

The code used to conduct this study can be found here: github.com/timbhiggins/Synoptic-Bench. The dataset can be found here: https://huggingface.co/datasets/Aikyam-Lab/Synoptic-Bench

Ethical Standards

The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

Author Contributions

T.H., A.M., and C.A. contributed new analytic metrics and wrote the manuscript. T.H. wrote all the codes, retrieved, processed, and harmonized datasets, and performed the analyses for technical validation of the new resource. A.M. and C.A. conceived the study.

References

[1] *Completing the Forecast: Characterizing and Communicating Uncertainty for Better Decisions Using Weather and Climate Forecasts*, National Academies Press, Washington, D.C., October 2006, Pages: 11699.

[2] *Valuing weather and climate: economic assessment of meteorological and hydrological services*, World Meteorological Organization, Geneva, Switzerland, 2015 (eng), OCLC: 912532797.

[3] Riley Brady and Aaron Spring, *climpred: Verification of weather and climate forecasts*, Journal of Open Source Software **6** (2021), no. 59, 2781.

[4] Jian Chen, Peilin Zhou, Yining Hua, Dading Chong, Meng Cao, Yaowei Li, Wei Chen, Bing Zhu, Junwei Liang, and Zixuan Yuan, *ClimateIQA: A New Dataset and Benchmark to Advance Vision-Language Models in Meteorology Anomalies Analysis*, Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (Toronto ON Canada), ACM, August 2025, pp. 5322–5333 (en).

[5] Xuming He, Zhiyuan You, Junchao Gong, Couhua Liu, Xiaoyu Yue, Peiqin Zhuang, Wenlong Zhang, and Lei Bai, *RadarQA: Multi-modal Quality Analysis of Weather Radar Forecasts*, 2025, Version Number: 1.

[6] Hans Hersbach, Bill Bell, Paul Berrisford, Shoji Hirahara, András Horányi, Joaquín Muñoz-Sabater, Julien Nicolas, Carole Peubey, Raluca Radu, Dinand Schepers, Adrian Simmons, Cornel Soci, Saleh Abdalla, Xavier Abellan, Gianpaolo Balsamo, Peter Bechtold, Gionata Biavati, Jean Bidlot, Massimo Bonavita, Giovanna De Chiara, Per Dahlgren, Dick Dee, Michail Diamantakis, Rossana Dragani, Johannes Flemming, Richard Forbes, Manuel Fuentes, Alan Geer, Leo Haimberger, Sean Healy, Robin J. Hogan, Elías Hólm, Marta Janisková, Sarah Keeley, Patrick Laloyaux, Philippe Lopez, Cristina Lupu, Gabor Radnoti, Patricia De Rosnay, Iryna Rozum, Freja Vamborg, Sebastien Villaume, and Jean-Noël Thépaut, *The ERA5 global reanalysis*, Quarterly Journal of the Royal Meteorological Society **146** (2020), no. 730, 1999–2049 (en).

[7] Himanshi Jain and Raksha Jain, *Big data in weather forecasting: Applications and challenges*, 2017 International Conference on Big Data Analytics and Computational Intelligence (ICBDAC) (Chirala, Andhra Pradesh, India), IEEE, March 2017, pp. 138–142.

[8] J.W. Jones, J.W. Hansen, F.S. Royce, and C.D. Messina, *Potential benefits of climate forecasting to agriculture*, Agriculture, Ecosystems & Environment **82** (2000), no. 1-3, 169–184 (en).

[9] Haobo Li, Zhaowei Wang, Jiachen Wang, Yueya Wang, Alexis Kai Hon Lau, and Huamin Qu, *CLLMate: A Multimodal Benchmark for Weather and Climate Events Forecasting*, 2024, Version Number: 2.

[10] Chengqian Ma, Zhanxiang Hua, Alexandra Anderson-Frey, Vikram Iyer, Xin Liu, and Lianhui Qin, *WeatherQA: Can Multimodal Language Models Reason about Severe Weather?*, 2024, Version Number: 2.

[11] Amanda M. Murphy, Cameron R. Homeyer, and Kiley Q. Allen, *Development and Investigation of GridRad-Severe, a Multiyear Severe Event Radar Dataset*, Monthly Weather Review **151** (2023), no. 9, 2257–2277.

[12] National Centers for Environmental Prediction/National Weather Service/NOAA/U.S. Department of Commerce, *NCEP GFS 0.25 Degree Global Forecast Grids Historical Archive*, 2015, Artwork Size: 574.507 Tbytes Pages: 574.507 Tbytes.

[13] C. A. Randles, A. M. Da Silva, V. Buchard, P. R. Colarco, A. Darmenov, R. Govindaraju, A. Smirnov, B. Holben, R. Ferrare, J. Hair, Y. Shinozuka, and C. J. Flynn, *The MERRA-2 Aerosol Reanalysis, 1980 Onward. Part I: System Description and Data Assimilation Evaluation*, Journal of Climate **30** (2017), no. 17, 6823–6850 (en).

[14] Shuo Tang, Jian Xu, Jiadong Zhang, Yi Chen, Qizhao Jin, Lingdong Shen, Chenglin Liu, and Shiming Xiang, *MeteorPred: A Meteorological Multimodal Large Model and Dataset for Severe Weather Event Prediction*, 2025, Version Number: 2.

[15] Louis W. Uccellini and John E. Ten Hoeve, *Evolving the National Weather Service to Build a Weather-Ready Nation: Connecting Observations, Forecasts, and Warnings to Decision-Makers through Impact-Based Decision Support Services*, Bulletin of the American Meteorological Society **100** (2019), no. 10, 1923–1942.

[16] Kingsley Eghonghon Ukhurebor, Charles Oluwaseun Adetunji, Olaniyan T. Olugbemi, W. Nwankwo, Akinola Samson Olayinka, C. Umezuruike, and Daniel Ingo Hefft, *Precision agriculture: Weather forecasting for future farming*, AI, Edge and IoT-based Smart Agriculture, Elsevier, 2022, pp. 101–121 (en).

[17] Sumanth Varambally, Marshall Fisher, Jas Thakker, Yiwei Chen, Zhirui Xia, Yasaman Jafari, Ruijia Niu, Manas Jain, Veeramakali Vignesh Manivannan, Zachary Novack, Luyu Han, Srikar Eranky, Salva Rühling Cachay, Taylor Berg-Kirkpatrick, Duncan Watson-Parris, Yi-An Ma, and Rose Yu, *Zephyrus: An Agentic Framework for Weather Science*, 2025, Version Number: 1.

[18] Mark Veillette, Siddharth Samsi, and Chris Mattioli, *SEVIR : A Storm Event Imagery Dataset for Deep Learning Applications in Radar and Satellite Meteorology*, Advances in Neural Information Processing Systems (H. Larochelle, M. Ranzato, R. Hadsell, M. F. Balcan, and H. Lin, eds.), vol. 33, Curran Associates, Inc., 2020, pp. 22009–22019.

Appendix

A. Experimental details

Implementation Details. We fine-tune each base model by training for roughly 1-3 epochs on one NVIDIA H200 GPU before running it on the test set. We use 2016-2022 for the training set, 2023 as the validation set, and 2024-2025 as the test set. The projector weights were fully fine-tuned, while Low-Rank Adaptation (LoRA) was used to train a small portion of the attention mechanism in the language model (<2%). We apply weight decay and dropout to the LoRA adapter branches to prevent overfitting. In addition, early stopping is used by computing Space-local scores for the validation set every 5000 steps.

Prompt Details. Base model and finetuned model versions for LLaVA-v1.5-7B, LLaVA-v1.5-13B, Qwen2-VL-7B, and LLaMA-3.2-11B-Vision were run on the test set. Each model uses the following prompt to generate discussions (see Fig. 1): "Analyze these weather charts showing maximum precipitation over the 3-day period (shaded contours), 500 mb geopotential height (unshaded contours) and 850

mb wind velocity (wind barbs) for the <[City, State]> forecast region and generate a detailed forecast discussion of the large-scale features relevant to the area. The forecast region is shown in the yellow box".

Additional Preprocessing Details. The images used in this study use the temporal mean of lead times ranging from 3 hours to 48 hours in 3 hour increments. The anomalies are calculated from monthly means ranging across the entire dataset. The month chose for climatology is always the same month that the AFD is issued. To create forecast images, 2-m temperature anomaly is capped between -5° and 5°C. 500 mb geopotential height contours are drawn at intervals of 60m and 850mb wind velocity barbs are plotted every 20th grid point to avoid visual clutter.

To filter AFDs, we only extract sentences containing the following keywords: *trough, trof, the low, this low, upper level low, low pressure, low-pressure, upper low, cyclone, closed low, cut-off low, troughing, ridge, the high, this high, upper level high, high pressure, high-pressure, upper high, anticyclone, blocking, ridging, cold front, warm front, warmer, cooler, and freezing*. We also exclude sentences with the following model keywords: *ECMWF, EURO, HRRR, ECCC, CMC, GEM, NAM, UKMET, ICON, RAP, SREF, HREF,* and synoptic pressure system keywords: *shortwave, short-wave, sfc trough, surface trough, sfc ridge, surface ridge, surface low, sfc low, surface high, sfc high* to increase the relevancy of NWS text in the use case presented in this study.

Parameter Settings. The same settings for the finetuning parameters were used for LLaVA-v1.5-7B, LLaVA-v1.5-13B, Qwen2-VL-7B, and LLaMA-3.2-11B-Vision. The learning rate was set at 2e-5, the batch size was set to 16, the LoRA rank was set to 8, and the LoRA alpha was set to 16. LoRA was applied to all linear modules within the attention and multi-layer perceptron blocks. AdamW was used as an optimizer. Image preprocessing was handled natively by each model's respective Hugging Face AutoProcessor. The images were converted to RGB and dynamically resized, padded, and normalized according to the base model's default vision encoder requirements (e.g., SigLIP for LLaVA/Llama and dynamic resolution for Qwen2.5-VL) before being tokenized.

Data Quality Control. Each text sample is matched to the closest GFS run. The initial raw dataset contained 1,367,041 samples. Text samples that are not within 12 hours of any GFS run are discarded. Pre-filtered samples in the raw dataset containing less than 30 words or greater than 200 words were also discarded. In addition, we removed samples containing text that suggested longer lead times beyond the 48-hour lead time post filtering (e.g. "Day 3", "extended period", "long term"). Quality control tests revealed zero samples containing text duplicates, unusual characters, absent image tokens, missing image files, or corrupted image files. Once all of the filtering specific to the example use case of this experiment was completed, the training set was reduced to 457,230 high-quality samples.

Extraction Details. We developed an automated Python pipeline to interface with the IEM JSON API, systematically querying and downloading historical AFDs from 117 NWS Weather Forecast Offices. The raw text files were extracted for the period spanning January 2019 to November 2025, yielding a comprehensive collection of chronologically ordered NWS discussions. To ensure strict temporal accuracy for multimodal pairing, we implemented a custom regular expression parser to extract the exact issuance timestamp directly from the raw text body of each product, converting all times to Coordinated Universal Time (UTC).

LLM as a Judge. Gemini-2.5-Flash was used as a judge for evaluation. The following text was used as the prompt for the judge: "Compare the generated forecast discussion to the ground truth forecast discussion. Evaluate how accurately the generated discussion captured the synoptic-scale features. Ignore minor differences in phrasing and focus on meteorological accuracy and completeness. Score the prediction from 0 to 1:

0: Completely incorrect or severely hallucinated.

.25: Major meteorological errors or missing critical synoptic features.

.5: Partially correct, but missing some context or containing minor errors.

.75: Mostly accurate and aligns well with the ground truth, minor omissions.

1: Excellent, highly accurate, and meteorologically sound compared to the ground truth.

Model	s_s^{loc}	s_m^{loc}	r_c^{loc}	s_s^{agg}	s_m^{agg}	r_c^{agg}
LLaVA-v1.5-7B-base	.0173$_{\pm.0028}$	.8278$_{\pm.0526}$	.0207$_{\pm.0031}$	.3447$_{\pm.0028}$	.5602$_{\pm.0526}$	.6017$_{\pm.0031}$
LLaVA-v1.5-13B-base	.0793$_{\pm.0048}$	**.9068**$_{\pm.0125}$	.0862$_{\pm.0051}$	.4229$_{\pm.0008}$	.6053$_{\pm.0056}$	.6843$_{\pm.0013}$
LLaMA-3.2-11B-Vision-base	.0671$_{\pm.0025}$	.7294$_{\pm.0075}$	.0886$_{\pm.0030}$	.3300$_{\pm.0025}$	.5683$_{\pm.0075}$	.5709$_{\pm.0030}$
Qwen2.5-VL-7B-base	.0491$_{\pm.0032}$	.8330$_{\pm.0155}$	.0577$_{\pm.0035}$	.3302$_{\pm.0032}$	.5478$_{\pm.0155}$	.5878$_{\pm.0035}$
LLaVA-v1.5-7B-LoRA	.1863$_{\pm.0056}$	.8839$_{\pm.0075}$	.2077$_{\pm.0060}$	**.4679**$_{\pm.0056}$	.5965$_{\pm.0075}$	.7718$_{\pm.0060}$
LLaVA-v1.5-13B-LoRA	.1862$_{\pm.0063}$	.8697$_{\pm.0085}$	.2098$_{\pm.0067}$	.4209$_{\pm.0063}$	.5870$_{\pm.0085}$	.7043$_{\pm.0067}$
LLaMA-3.2-11B-Vision-LoRA	.1726$_{\pm.0055}$	.8992$_{\pm.0073}$	.1892$_{\pm.0057}$	.4597$_{\pm.0055}$	**.6124**$_{\pm.0073}$	.7395$_{\pm.0057}$
Qwen2.5-VL-7B-LoRA	**.1886**$_{\pm.0074}$	.8660$_{\pm.0058}$	**.2128**$_{\pm.0062}$	.3992$_{\pm.0074}$	.5978$_{\pm.0058}$	.6529$_{\pm.0062}$
Gemini-3.1-Pro	.0397$_{\pm.0050}$	.7725$_{\pm.0340}$	.0408$_{\pm.0057}$	.3963$_{\pm.0050}$	.5135$_{\pm.0340}$	**.7732**$_{\pm.0057}$
Nearest Neighbor	.0585$_{\pm.0036}$	.7532$_{\pm.0166}$	.0753$_{\pm.0041}$	.3216$_{\pm.0036}$	.5607$_{\pm.0166}$	.5614$_{\pm.0041}$
Climatology	.0850$_{\pm.0042}$	.7728$_{\pm.0148}$	.1073$_{\pm.0048}$	.4042$_{\pm.0042}$	.5795$_{\pm.0148}$	.6894$_{\pm.0048}$
Llama-3.2 (blind)	.0912$_{\pm.0048}$	.8664$_{\pm.0141}$	.1043$_{\pm.0051}$	.4211$_{\pm.0048}$	.5885$_{\pm.0141}$	.7050$_{\pm.0051}$

Table 4. *SPACE was used for synoptic temperature system evaluation for base models, finetuned models, and baselines. We calculate the mean and standard error of the SPACE-local and SPACE-aggregate scores (s_s), match scores (s_m), and coverage ratios (r_c) within the test set. The range of possible scores for all metrics is 0 to 1, with 1 being a perfect score.*

Location Stop Nodes. The following locations are excluded from functioning as common relatives or matching to other locations via a common relative: "Canada", "CONUS", "Eastern Canada", "Central Canada", "Western Canada", "Eastern CONUS", "Western CONUS", "Central CONUS", "Central Plains", "Ohio Valley", "Great Lakes", "Central U.S.", "Eastern U.S.", "Western U.S.", "Central United States", "Eastern United States", "Western United States", "Central U", "Eastern U", "Western U", "Eastern US", "Western US", "Central US", "Midwest", "The Plains".

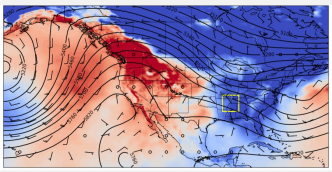
Space Temperature Scores. SPACE scores for temperature are shown in Table 4. The following key terms for temperature polarity were used: "warm front", "warmer than average", "warmer than normal", "above average temperature", "cold front", "cooler than average", "cooler than normal", "colder than normal", and "below average temperature". The number of temperature terms found in both AFDs and generated text was dominated by "cold front" and "warm front" (>80%). Fine-tuned Qwen2.5-VL-7B-LoRA had the highest Space-local scores while fine-tuned LLaVA-v1.5-7B had the highest Space-aggregate scores. The 48-hour temporal average that was used to create the images could lead to too much smoothing for the temporal scale at which fronts often take effect. Although Space scores could be effective for synoptic temperature systems, altering training images and experimental setup to target temperature-related phenomena would improve the reliability of evaluation in future work.

Additional Example Cases. Three additional example cases are shown in Figures A1-A3. In Case 3, there are two matching instances of pressure systems and two unmatching instances of pressure systems, leading to a coverage ratio of 0.5. Of the matching instances, both indicated high pressure, resulting in a match score of 1 and a SPACE-local score of 0.5. In Case 4, the prediction described synoptic pressure systems, while the reference text did not (Short wave troughs are not considered synoptic pressure systems). This results in a SPACE-local score that defaults to 0. Although there was some accurate information in the generated text, it was not relevant enough to the area to be included in the AFD. The pressure systems described in the generated text could still impact SPACE-aggregate scores. In Case 5, all pressure systems were described in the same general area, leading to a coverage ratio of 1. The match score was imperfect due to a ridge that would impact the region soon after the trough.

Training Set Size. The impact of the size of the training set on the Space scores is shown in Figure A4. The majority of the improvement in Space scores occurs within the first half epoch of training.

Case 3:

Place: Memphis, TN; Date: 02/09/15
s_s = .5; s_m = 1; r_c = .5
F1 = 0.2208; Bertscore = 0.8001
METEOR = 0.1558; ROUGE-L = 0.1618



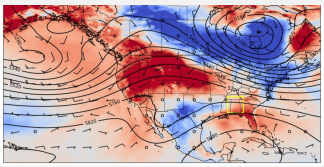
Reference: AT 3 PM SURFACE ANALYSIS DEPICTED A **LOW PRESSURE** CENTER OVER **CENTRAL NORTH CAROLINA** WITH A COLD FRONT TRAILING FROM THE LOW CENTER SOUTHWESTWARD ALONG THE GULF COAST. CLOUDS CONTINUE TO FILTER INTO THE REGION WITH NORTHERLY WINDS IN THE WAKE OF THE AFOREMENTIONED COLD FRONT EARLIER TODAY. COOL AND DRY CONDITIONS ARE THE HIGHLIGHT OF THE FORECAST PACKAGE THIS AFTERNOON WITH BELOW NORMAL TEMPERATURES RETURNING THIS WEEKEND. THE **UPPER LEVEL TROUGH** LOCATED OFF TO **OUR EAST** WILL CONTINUE ITS EASTWARD TRANSIT OVERNIGHT. AND THUS CLOUD COVER WILL DIMINISH ON TUESDAY AFTERNOON AS **HIGH PRESSURE** BUILDS IN AT THE SURFACE WITH **RIDGING ALOFT**. WINDS WILL PERSIST FROM THE NORTH ON TUESDAY KEEPING HIGHS COOLER THAN MONDAY.

Prediction: The weather in Memphis on Tuesday will be mild and dry. A **high pressure ridge** will remain in place over **the region**, allowing for warm temperatures and light winds. The weather will be even warmer in the eastern parts of the country, where temperatures will be in the 70s.

Figure A1. *Example Case 3.*

Case 4:

Place: Birmingham, AL; Date: 03/20/15
s_s = 0; s_m = 0; r_c = 0
F1 = 0.0667; Bertscore = 0.8250
METEOR = 0.0779; ROUGE-L = 0.0811



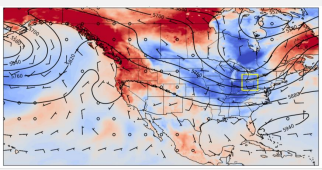
Reference: WE HAVE A SOUTHERN STREAM SHORT WAVE TROUGH MOVING ACROSS THE AREA.

Prediction: As the upper level **trough** moves to **the east**, the low level flow will shift to the east-southeast. The surface flow will be southerly over the next couple of days, with a **low pressure** area developing over the **eastern Gulf of Mexico**. **High pressure** will build over the **western United States**. A **ridge** of **high pressure** will build over **the Southeast**.

Figure A2. *Example Case 4.*

Case 5:

Place: Cleveland, OH; Date: 08/03/17
s_s = .67; s_m = .67; r_c = 1
F1 = 0.1942; Bertscore = 0.8250
METEOR = 0.1253; ROUGE-L = 0.2149



Reference: **Low pressure** from the upper **midwest** will move across the **Great lakes** Friday pulling a strong cold front east across the area Friday. **High pressure** will build across **the region** over the weekend.

Prediction: The latest forecast discussion for the large-scale features indicates that a **low pressure** system will continue to develop and track across the eastern **Great Lakes** this afternoon and evening. **This low** will deepen and intensify as it moves over the colder waters of **Lake Superior** tonight. **The low** will bring gusty winds and rain to **the region**, with a potential for thunderstorms and heavy rain. The winds will be strongest in the morning and afternoon hours, with gusts over 40 mph expected across the eastern Great Lakes.

Figure A3. *Example Case 5.*

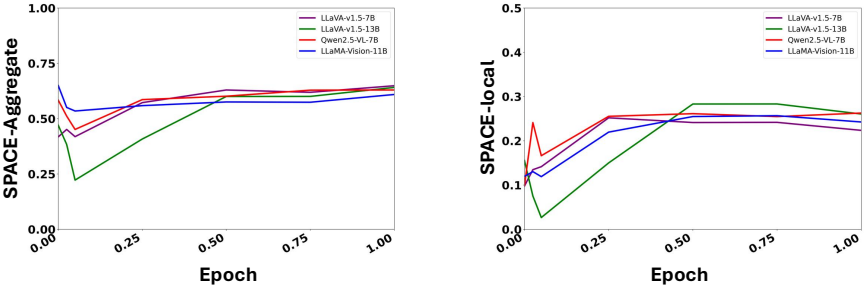


Figure A4. *Training Set Size.*